Investigation of Non-Convolutional VAE Decoders

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Abstract

The use of convolutional decoders in VAEs has become the standard in many applications, especially in tasks like image generation, because convolutional networks are particularly well-suited to processing image data, which has a spatial structure that convolutions can effectively leverage. However, this investigation seeks to discover if non-convolutional decoders may also be effective at decoding VAE-generated images and, if so, to what degree they are capable of doing so in comparison to traditional convolutional counterparts.

1. Project Overview

1.1. Inspiration

Much of designing Neural Network architecture is currently based on historical patterns of success such that the design of Neural Network architectures is often influenced by historical successes, leading to designs that closely follow previous models built for similar tasks.

However, given the experimental nature of building out Neural Networks and the drastic differences minor differences in architecture can yield, independent investigation of alternatives to the assumed standards has the potential to lead to discovering methods that may better suit a designer's unique needs.

That being the case, while the general reasoning and corresponding consensus of using convolutional nodes in tasks such as image generation makes reasonable sense, a personal investigation into the potential of non-convolutional methods serves to ensure that no potential alternative model of architecture design goes overlooked simply due to following the status quo.

1.2. Method

This investigation compares three main components:

1. A traditional VAE Convolutional Decoder

2. A simple Non-Convolutional Multi-Layer Perceptron Decoder

3. A modified Non-Convolutional SIREN Decoder

All three are trained on, and passed, the resulting latent vectors from a standard Variational Autoencoder. The results are then visualized and compared, identifying to what degree the non-convolutional decoders might act as meaningful alternatives to the convolutional decoder. Further discussion follows, recapping the findings and evaluating the impact of this investigation.

2. Experimental Procedure

2.1. Training the VAE and Convolutional Decoder

This series of experiments begins by setting up our baseline generative model. We use a standard Variational Autoencoder (VAE) which allows us to create latent representations of data as well as reconstruct it through its corresponding convolutional decoder. This convolution decoder also acts to establish a foundation for comparing different decoder architectures and their ability to capture complex patterns in the data.

The constructed VAE was trained on Fashion-MNIST, a collection of 28x28 pixel images of various types of clothing. This dataset was intentionally selected as its relatively simple nature both:

A.) allows it to act as a general proof of concept

B.) allows for faster iteration and testing which is essential given the experimental nature of this investigation.

2.2. Training the Non-Convolutional Decoders

After the VAE is trained, we begin by training our basic Multi-Layer Perceptron by providing it a sample latent vector representation and then directly comparing it to the VAE Reconstruction as generated by the VAE's convolutional decoder.

This training structure was specifically designed in order to train additional decoders on the same input as the convolution decoder rather than the decoded images themselves. This in turn allows for each decoder to remain independent and distinct which allows for direct comparison.

At this point there emerged a curiosity as to how another, more complex, non-convolutional structure might compare as a possible decoder: the SIREN model as seen in "Implicit

Neural Representations with Periodic Activation Functions" (Vincent Sitzmann et al., 2020).

While it is true that the SIREN model was designed for implicit representations, its use of Sinusoid activation functions provided a unique, new non-convolutional structural possibility that may yield an alternative replacement decoder.

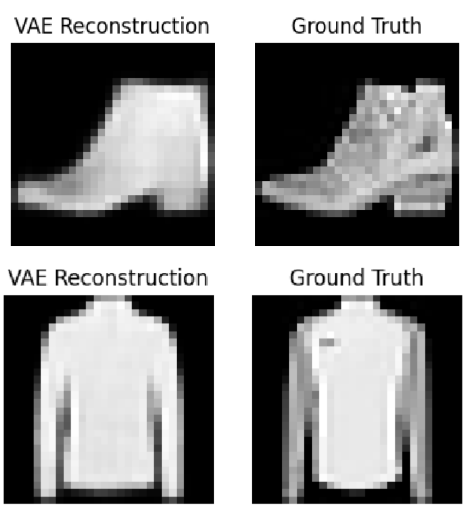
In order to test this possibility, the basic SIREN model discussed in Vincent Sitzmann et al. was integrated along with an adjustment that allowed it to take in latent vectors as an argument such that it could now be trained on the same input as the other decoders.

From there it was trained in the same fashion as the MLP, albeit with a slightly lower learning rate. This was because through experimentation it was found that the more complex SIREN architecture seemed to benefit from a more consistent path of improvement achieved through the lower rate.

3. Results

3.1. Convolutional Decoder Results

When reviewing the results, the VAE convolutional decoder generally did a decent job at capturing the forms of both simple and complex items such as the shoe and long-sleeve shirt seen below, although with much of the fine details lost to abstraction and encoding.

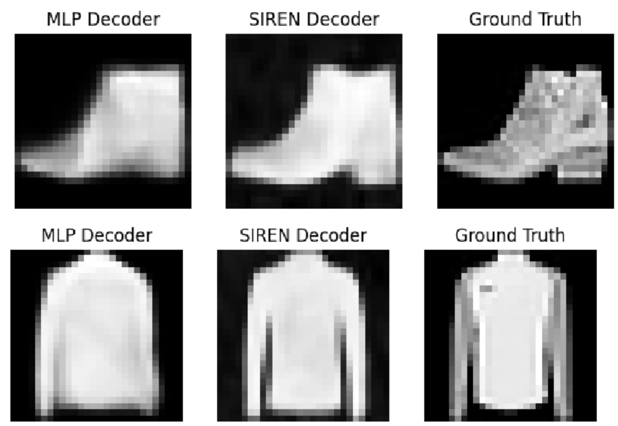


These results are relatively unsurprising given similar convolutional models being the de facto standard for VAE decoders, but in the case of this investigation specifically these outputs help to at least reconfirm that convolutional decoders are indeed a valid choice for image generation in general.

3.2. Non-Convolutional Decoder Results

This section presents the results of the non-convolutional decoders, including both the MLP and SIREN models.

While the MLP Decoder was relatively successful in decoding simple forms such as the shoe, it struggled as the complexity of the image increased. As a result it was not uncommon to see the MLP output decoded images that lacked more complicated details such as gaps/cutouts seen in the ground truth, like as the between the sleeves and body of the shirt below.



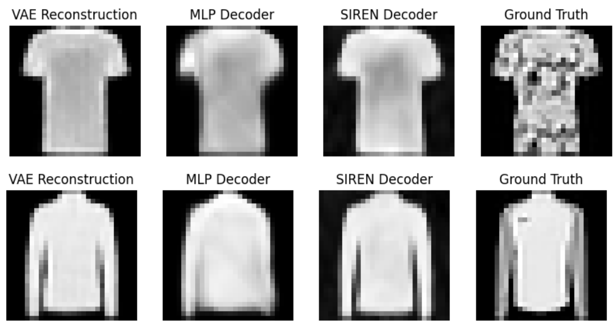
However, the SIREN model showed surprising levels of accuracy in decoding images in that its output often closely mirrored that of the convolutional VAE decoder as well as the Ground Truth itself. This result was commonly seen in generations of both simple and complex clothing pieces. These consistently accurate and detailed generations seem to imply it may potentially be a possible alternative to convolutional decoders.

Of note, aside from architecture, the only difference in the training cycles for the MLP and SIREN models was their respective learning rates. Experimental trials revealed that the MLP model worked well with a learning rate of .001 while the SIREN model worked best with a lower learning rate of .0005.

Both were trained with twenty epochs and a batch size of one-hundred twenty-eight and used the exact same iteration of the trained VAE in their training comparisons, making the difference in their results almost certainly due to their respective architecture designs.

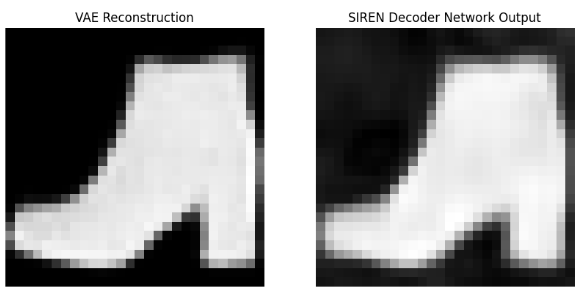
3.3. Discussion

While the MLP model clearly struggled to decode more complex images, the SIREN model fared much better to the point that this investigation revealed there may in fact be a world in which the modified SIREN model could be used as an alternative, non-convolutional decoding structure.



Note the VAE and SIREN Decoder output similarities above. As seen in the figures, the VAE and SIREN decoder outputs are highly similar, indicating that the SIREN decoder is indeed effective in reconstructing the images. Even as the MLP model blurs, the SIREN Decoder shows it is capable of accurately representing the encoded form even without the use of convolutional nodes.

Looking closer we can further directly compare the modified SIREN to the convolutional VAE decoder:



The incredible similarities seen in this result, in tandem with the previous figures, reveal that the SIREN Decoder is not only capable of accurately decoding the VAE encodings, but in fact can do so at very high level of accuracy.

While further testing is required to investigate how these findings scale with image complexity, these results reveal with a non-insignificant degree of certainty that there exists cases where non-convolutional models can in fact act as VAE decoders while maintaining a high level of precision.

Not to be mistaken, this paper does not claim in any way that these results prove non-convolutional decoders are more efficient or superior to convolution decoders - rather it simply serves as evidence of the existence of more options in architecture design than strictly using convolutional methods.

In doing so it successfully reaffirms that in the search for optimal Neural Network architecture, one isn't limited to the current design status quo and that, while there are often fundamentally sound principles behind current best practices, they should not limit a designer's creativity in exploring new potential alternatives to assumed solutions.

References

Sitzmann, V., Martel, J., Bergman, A., Lindell, D., & Wetzstein, G. (2020). Implicit neural representations with periodic activation functions. Advances in neural information processing systems, 33, 7462-7473.